Utilizing Machine Learning to Classify Type of Heartbeat Based on Electrocardiogram Data

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My Story

- Collegiate cross country runner
- 2021: Symptoms arise
- 2022: Heart episode during a run
- Initial Diagnosis: Arrhythmogenic Right Ventricular Cardiomyopathy (ARVC)
- Correct Diagnosis: McLeod Syndrome

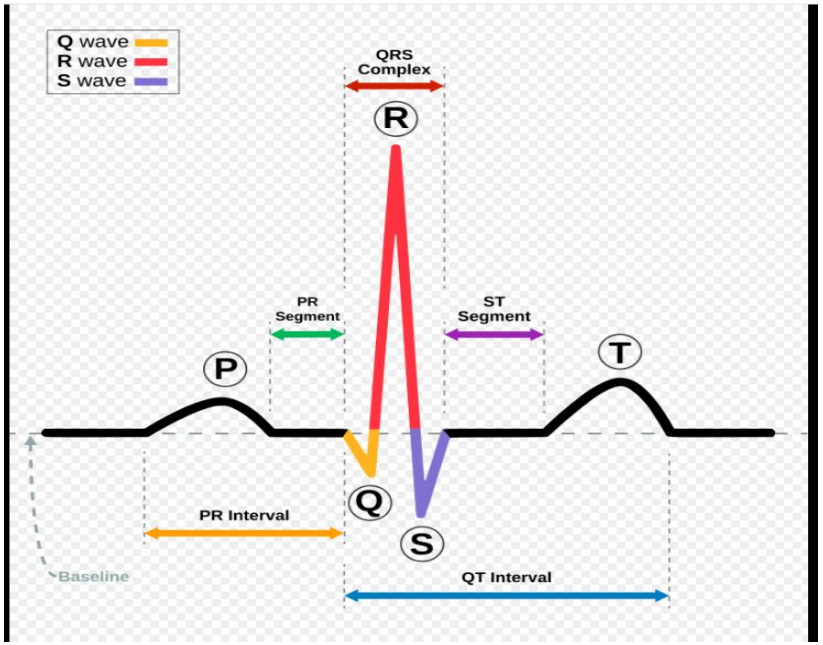
Running Journey

My Question

How can I use my experiences and education to help athletes detect underlying heart issues before they become life-threatening?

What is an Electrocardiogram?

- An electrocardiogram (ECG) is a test that checks the heartbeat and records electrical signals to determine irregular heartbeats called arrhythmias
- The test is performed by attaching electrodes on the chest and arms of the patients
- The standard type of ECG is a 12-lead test that provide 12 distinct angles of the heart



What is a Cardiac Arrhythmia?

- Cardiac arrhythmias are an underlying sign of dangerous heart conditions
- They cause the heart to not pump blood adequately to the rest of the body
- Symptoms include fainting, dizziness, weakness, and fast heart palpitations

Proposition

SAVE ATHLETES' LIVES THROUGH ML-POWERED CARDIAC SCREENING

I propose integrating ECG testing with machine learning-based heartbeat classification in pre-season physicals for all athletes. This approach can help trainers and physicians identify abnormal cardiac patterns, enabling timely referrals and earlier intervention.

Where We Stand Today

Current Status

Certain leagues and states require ECG testing for athletes

Problem

Many physicians/trainers lack expertise in athletic ECG's -> Unnecessary referrals or premature clearance to play

Potential Solution

AI/ML models for accurate ECG interpretation

Dataset

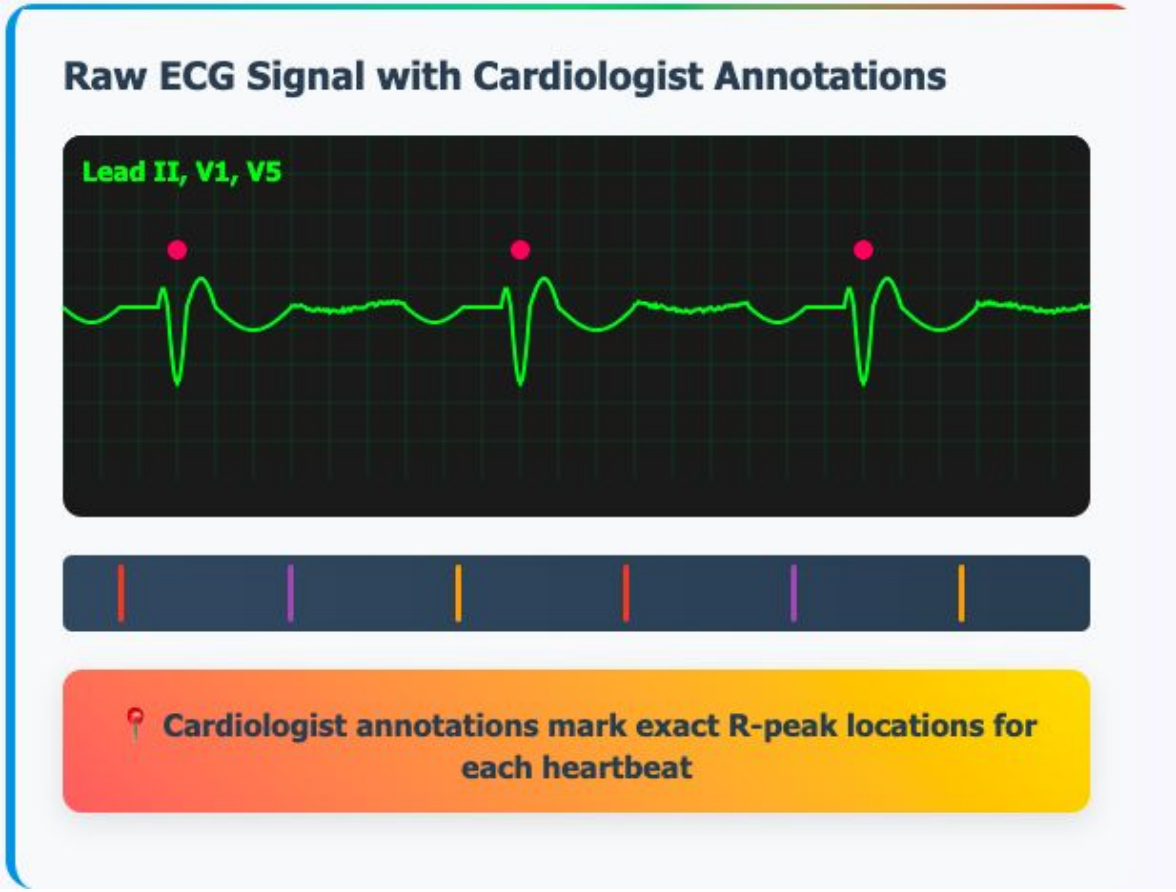
- **Source**: St. Petersburg Institute of Cardiological Technics (INCART) database, accessed via WFDB (PhysioNet)
- **Annotations**: Cardiologist-labeled heartbeat types
- **Size**: 75 half-hour, 12-lead ECG recordings
- **Patients**: 32 individuals
- **Records**: 175,560 raw signals

Target Variable - Heartbeat Type

Heartbeat Type - AAMI Standard	**Description**
Normal (N)	Normal heartbeat
Supraventricular Ectopic Beat (SVEB)	Abnormal beat originating in the atria that can lead to longer term risk of atrial fibrillation or stroke
Ventricular Ectopic Beat (VEB)	Abnormal beat originating in the ventricles that can trigger immediate life-threatening arrhythmias

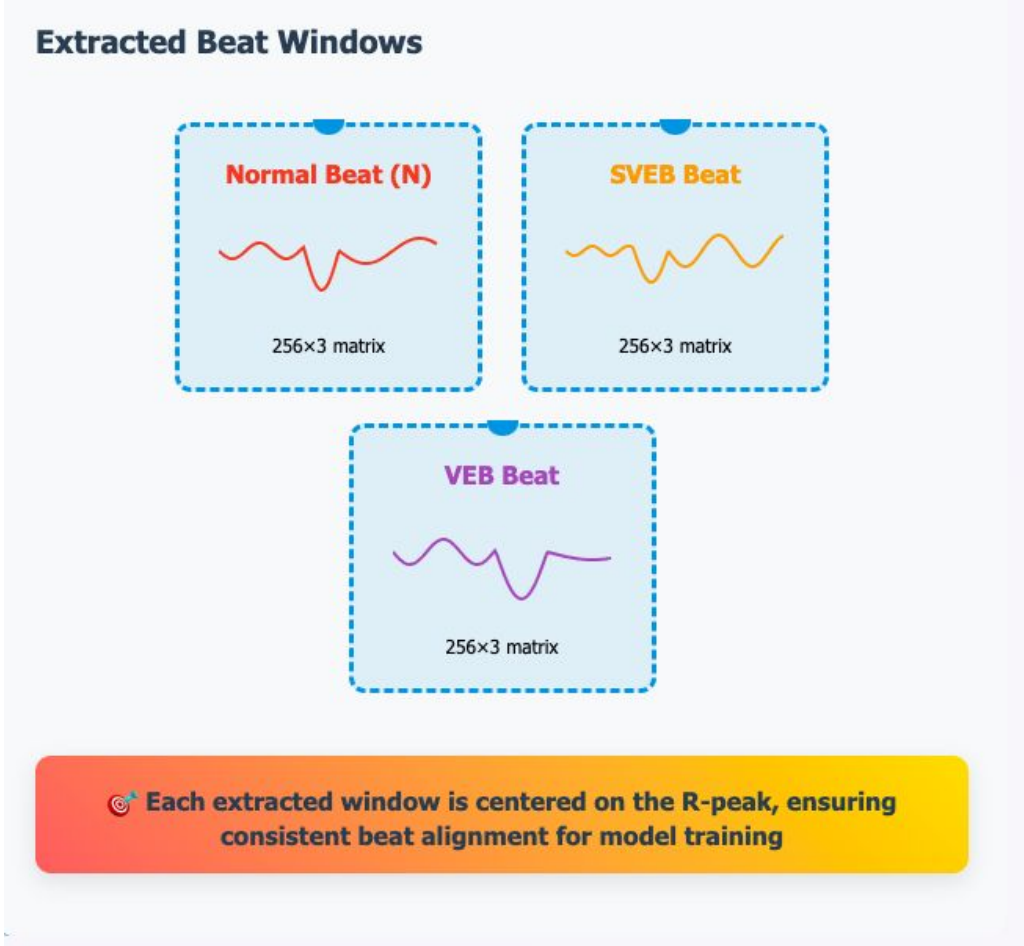
Signal Processing Pipeline: Beat Detection

- Raw ECG Signal → **Beat Detection**



Signal Processing Pipeline: Window Extraction

- Raw ECG Signal → Beat Detection → **Window Extraction**



Signal Processing Pipeline

- **Processing**
  - Raw ECG Signal → Beat Detection → Window Extraction → **Normalization** → **ML Model**

- **Normalization**
  - Applied per-lead Z-score transformation to standardize amplitude across windows

- **Learning Approach**
  - Processes raw ECG signals directly into the model without the need for human feature engineering

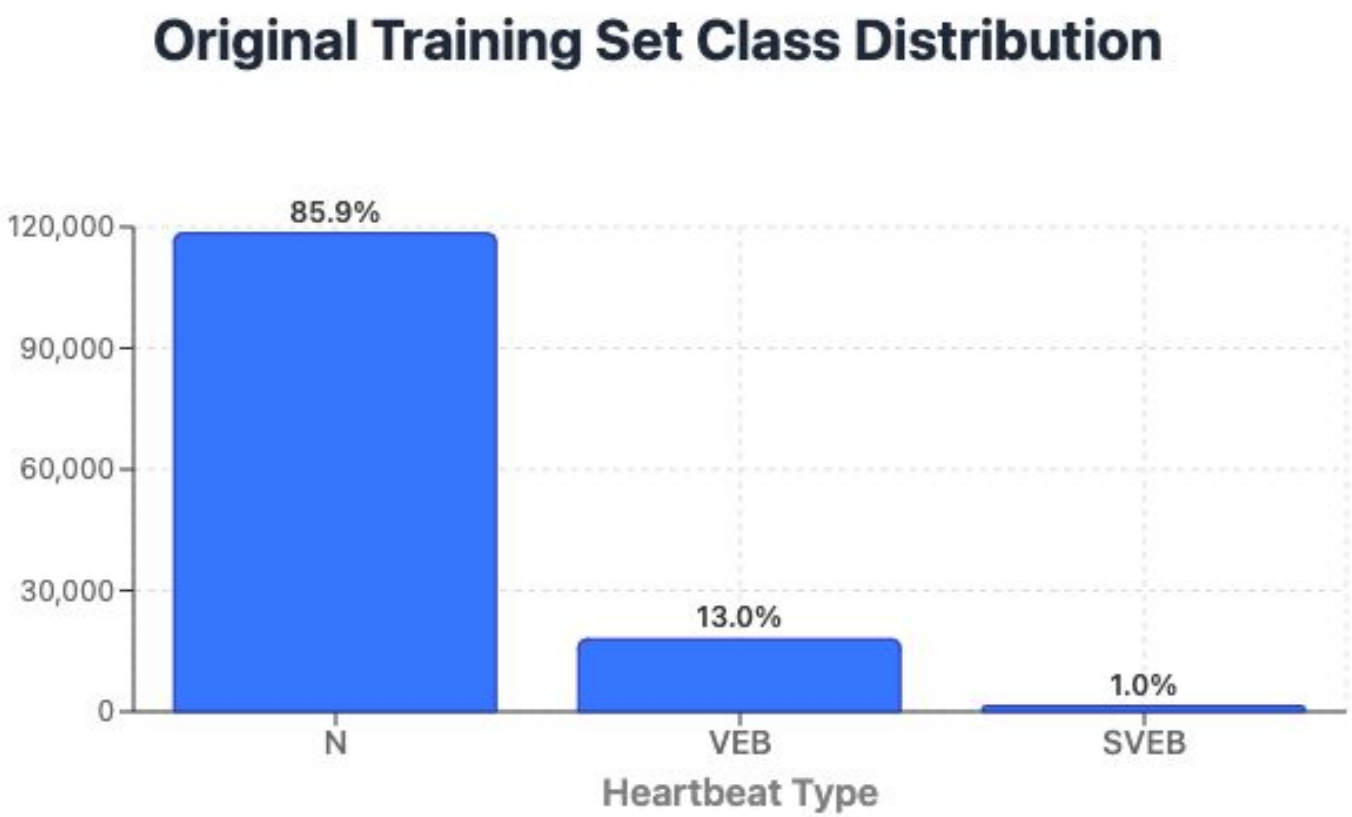
Patient-Wise Split

- **The Goal:** Train the model to learn heartbeats, not specific patients

- **The Problem:** A standard random split can lead to data leakage

- **The Solution**: I used a patient-wise split with GroupShuffleSplit

Class Distribution



4 Stage Approach to Fix Imbalance

Philosophy: Hybrid approach combining multiple techniques

1. **Under-sampling:** Reduce computational burden
2. **Borderline SMOTE:** Generate synthetic minority samples
3. **Domain-aware augmentation:** Make synthetic beats more realistic
4. **Cleaning**: Remove noisy boundaries

Goal: Create learnable distribution while preserving medical realism

Stage 1: Undersampling N

- **Method**
  - Use RandomUnderSampler in Python

- **Decision**
  - N Beats: Reduce by **50%**

- **Rationale**
  - Maintains clinical reality (Normal beats should dominate)
  - Reduces computational complexity

Stage 2: Oversampling

- **Challenge**
  - Extreme class imbalance prevents effective learning of rare but clinically critical arrhythmias

- **Decision**
  - SVEB: Oversample by factor of **20**
  - VEB: Oversample by a factor of **2**
  - Method: BorderLine SMOTE

- **Rationale**
  - Helps the model learn robust patterns

Stage 3: Domain-Aware Augmentation

- **Techniques Applied**
  - Gaussian noise: $\sigma = 0.05$ (realistic ECG noise)
  - Gaussian smoothing: $\sigma = 0.5$ (removes unwanted high frequency signals)
  - Time warping: ±5% stretch/compression for SVEB and ±2% for VEB

- **Result**
  - Synthetic samples become more realistic and diverse through post-SMOTE augmentation

Stage 4: Cleaning and Final Result

- **Strategy**
  - Use Tomek Links to remove ambiguous boundary samples
- **Impact**
  - N class reduced by 10 samples
- **Final Result**

Heartbeat Class	Training Samples (Before)	Training Samples (After)
N	118,452	59,219
SVEB	1,418	28,360
VEB	17,950	35,900
Total	137,820	123,479

Convolutional Neural Network (CNN)

- **Definition**
  - A CNN is a type of deep learning model designed to process grid-like data by utilizing layers

    - Convolutional layers detect features using filters
    - Pooling layers reduce data size
    - Fully connected layers classify the features

- **Why It Works for My Data**
  - CNN learns features directly from raw signals, bypassing the need for hand-crafted features

CNN Architecture Design

5-Layer CNN with CBAM Attention & Residual Learning

Input (256×3) → 5× Conv1D + CBAM + Residual →

Dense + Softmax (3 classes)

Core Architecture

- **5 Conv1D Layers** [64, 128, 256, 256, 512] filters
- **3 Residual Blocks** for gradient flow
- **30% Dropout** + Batch Normalization

CBAM Attention Mechanism

- **Channel Attention** (ratio=8) + **Spatial Attention** (k=7)
- **Applied After Each Conv** for feature enhancement
- **Hyperparameter Optimization** via Optuna

Focal Loss and Training Strategy

Focal Loss Function

Custom focal loss with **aggressive focus on minority class** (Gamma=4.0) and asymmetric class weighting

Impact: Addresses severe class imbalance by down-weighting easy examples

Advanced Training Strategy

F1-score monitoring with early stopping for medical classification

Optuna optimization: Automated hyperparameter tuning

Ensemble learning: 3-model ensemble with diverse seeds

Extended patience: 120 epochs to handle ECG complexity

Ensemble Learning

3-Model Ensemble with Diverse Training

Ensemble Strategy

- **Different Seeds** : 42, 52, 62 for diverse initialization
- **Independent Training** : Each model sees different augmented data
- **Probability Averaging** : Mean of softmax outputs across models
- **Best Hyperparameters** : All models use Optuna-optimized settings

Diversity Sources

- **Weight Initialization** : Random starting points create different learning paths
- **Augmentation Variance** : Each model sees different noise, warping, shifts
- **Training Variance** : Different validation splits and convergence paths
- **Error Compensation** : Individual weaknesses balanced by ensemble strength

Model Performance Evaluation Metrics

- Recall answers: Of all the actual abnormal heartbeats, how many did the model correctly identify?

- Precision answers: Of all the beats the model flagged as abnormal, how many were actually abnormal?

- F1-score, a harmonic mean of precision and recall, serves as the primary metric for model evaluation

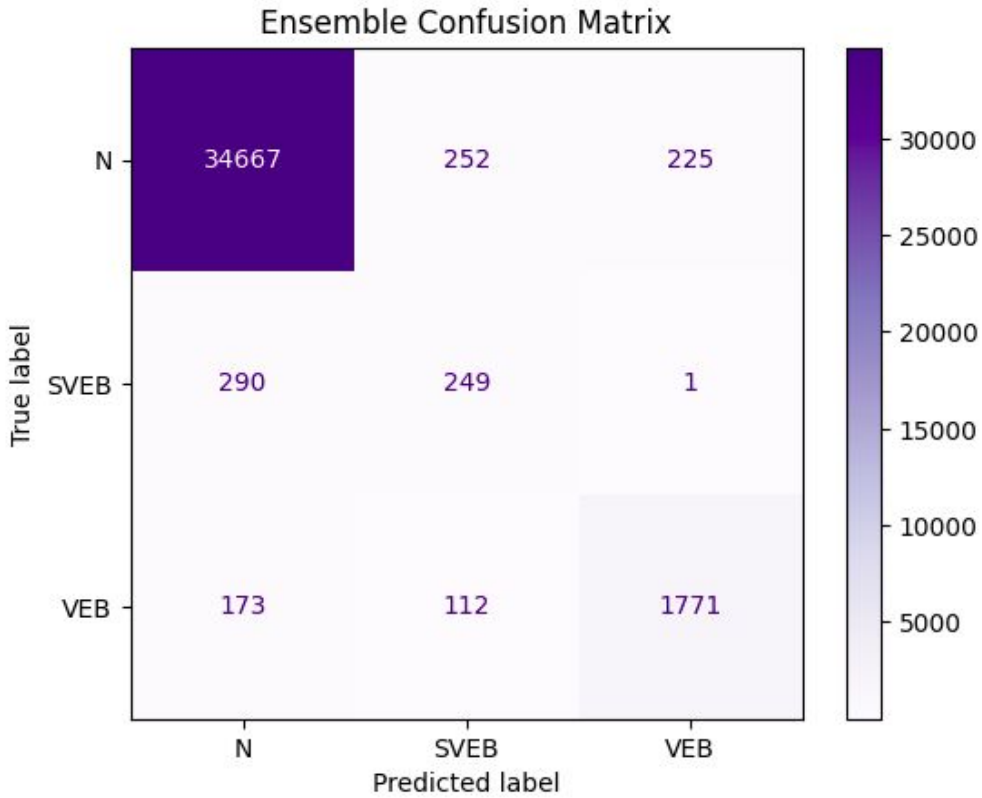
CNN Classification Report

- High overall accuracy of 0.97
- Good Macro Average F1 of 0.76
- Excellent performance on N and solid performance for VEB
- SVEB F1 score is poor with low precision and recall, due to rarity

Class	Precision	Recall	F1-Score
N	0.99	0.99	0.99
SVEB	0.41	0.46	0.43
VEB	0.89	0.86	0.87

Metric	Score
Accuracy	0.97
Macro Average F1	**0.76**

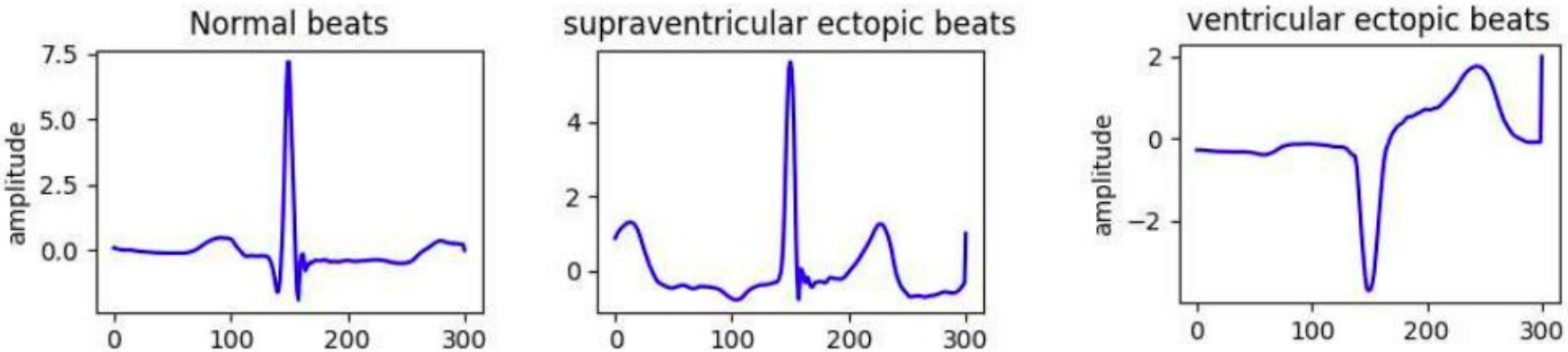
Confusion Matrix



- Normal: Excellent performance with minimal misclassifications
- VEB: Strong performance with small number of misclassifications
- SVEB: Correctly identified fewer than half the total SVEB cases with 290 false negatives misclassified as Normal

Discussion: SVEB

- **Technical Challenges**
  - SVEB closely resemble N beats
  - Severe class imbalance compounds detection difficulty



Discussion Clinical Significance

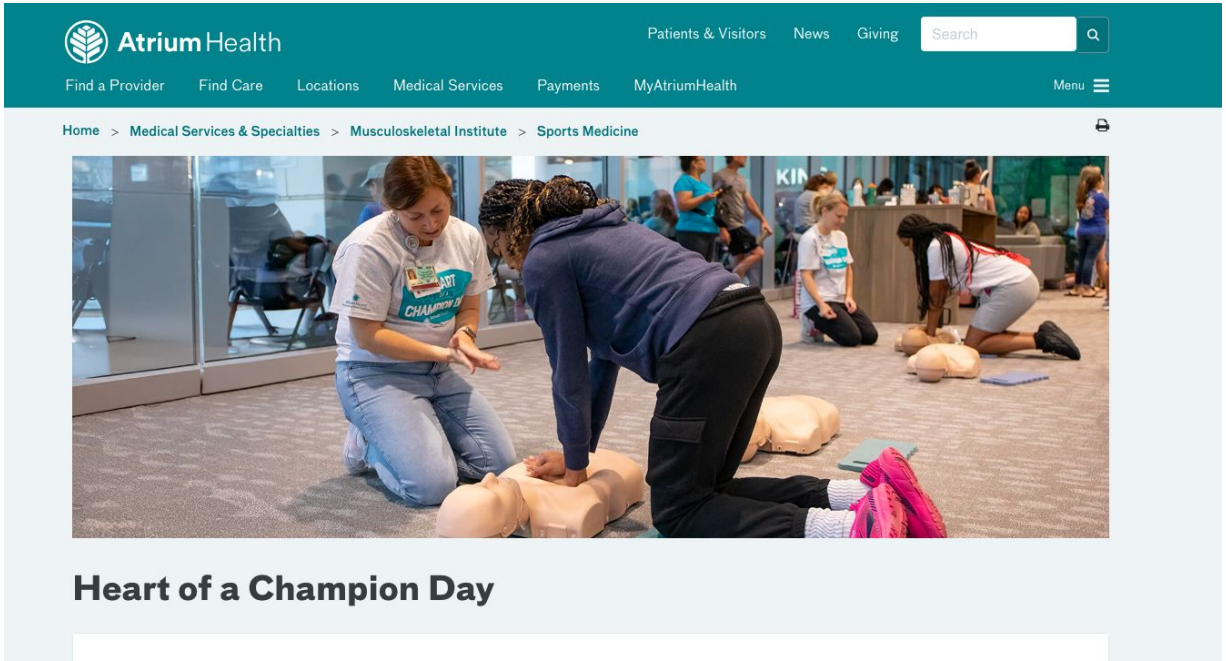
- **Clinical Significance**
  - VEB and SVEB require detection in athletic screening
  - VEB poses immediate risk requiring urgent evaluation
  - SVEB indicates potential long-term stroke/atrial fibrillation risk
- **Model Performance Impact**
  - Strong N and VEB classification provides reliable normal beats and life-threatening arrhythmias detection
  - Poor SVEB performance requires future methodological improvements

Limitations

- **Dataset Limitations**
  - INCART data has severe class imbalance
- **Generalization Constraints**
  - Limited population diversity of 32 patients
- **Clinical Translation Gap**
  - Unable to validate against real clinical outcomes
- **Synthetic Data Limitations**
  - BorderlineSMOTE does not capture true physiological variability
  - Requires knowledge of electrophysiological mechanisms and inter-patient variability in cardiac anatomy

Future Directions

- <u>Heart of a Champion</u> screens Charlotte high school athletes
  - Presents an opportunity to train and test model
- Generate open source datasets of athletes' ECGs to build and test models



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